

EMCH 260

SOLID MECHANICS

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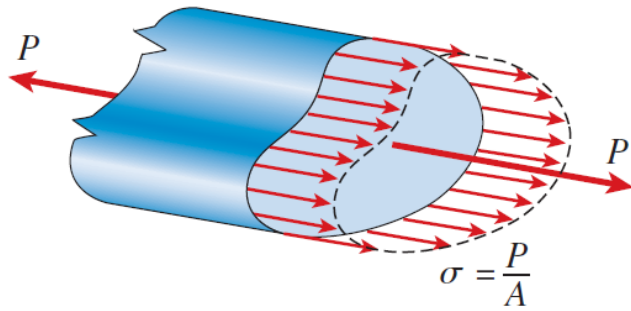
Stress

1

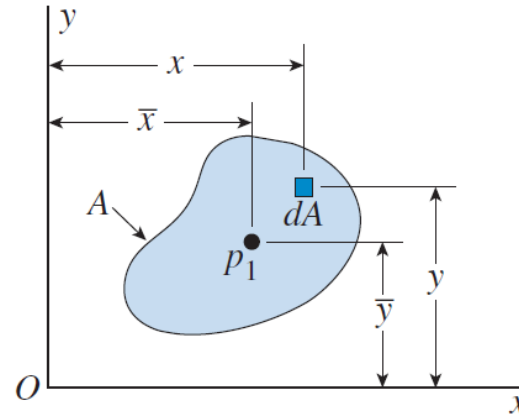
Chapter 1.4

- ✓ Analyze and design of members subjected on axial load or shear
- ✓ Know how to do allowable stress design

AVERAGE NORMAL STRESS AND SHEAR STRESS

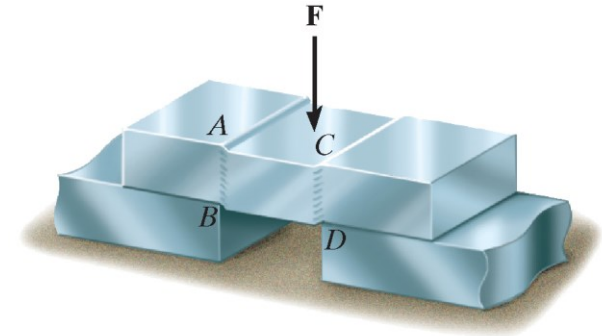


(a)

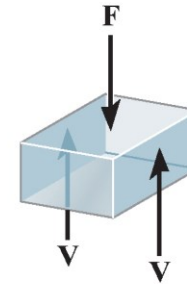


(b)

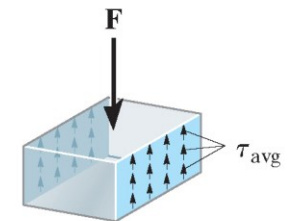
$$\sigma_{\text{avg}} = \frac{P}{A}$$



(a)



(b)



(c)

$$\tau_{\text{avg}} = \frac{V}{A}$$

DESIGN OF SIMPLE CONNECTION

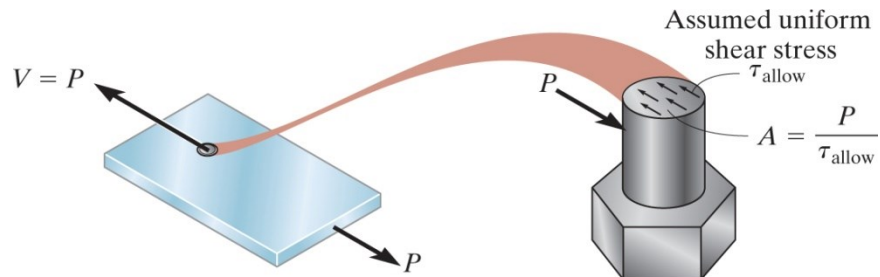
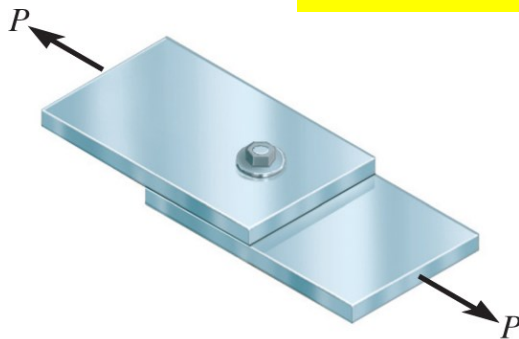
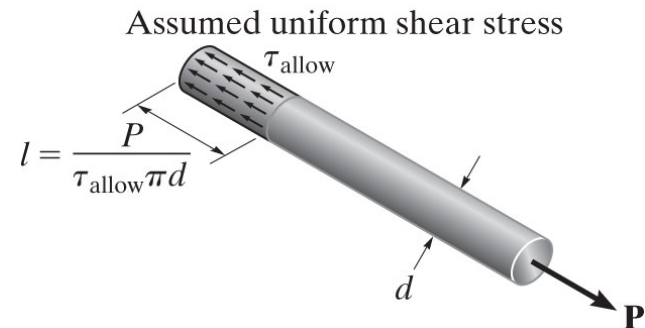
- For normal force requirement

$$A = \frac{P}{\sigma_{allow}}$$



- For shear force requirement

$$A = \frac{V}{\sigma_{allow}}$$



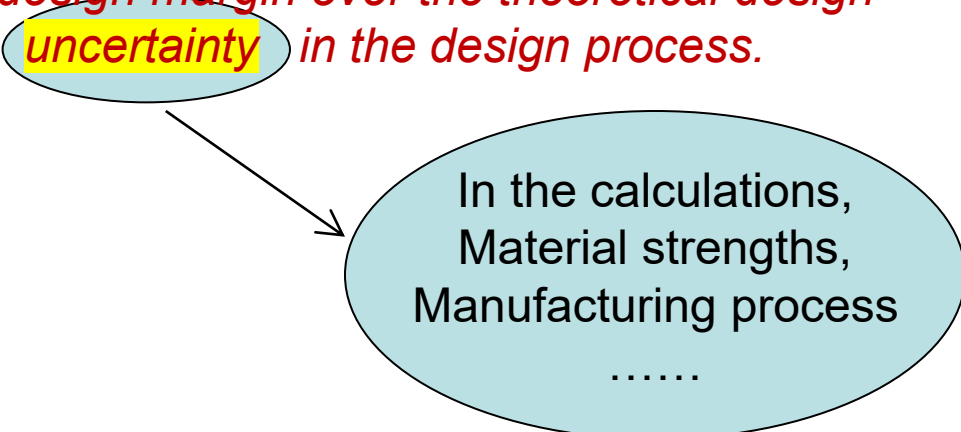
ALLOWABLE STRESS DESIGN

- **Factor of Safety (F.S.):** ratio of the failure load to the allowable load

$$F. S. = \frac{F_{fail}}{F_{allow}}$$

E.g.: If a component needs to withstand a load of 100 N and a F.S. of 3 is selected then it is designed with strength to support 300 N.

- ✓ *The larger the F.S., the more safety a structure has*
- ✓ *Is used to provide a design margin over the theoretical design capacity to allow for uncertainty in the design process.*

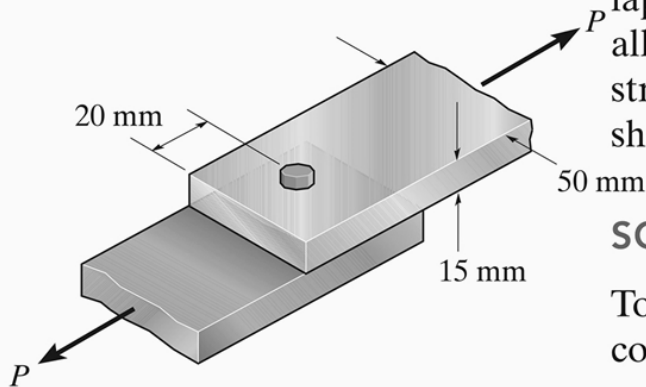


In the calculations,
Material strengths,
Manufacturing process
.....

FACTOR OF SAFETY

F.S. (<i>Based on yield strength</i>)	Applications
1.25 – 1.5	Material properties known in detail / Operating conditions known in detail / Load and the resulting stresses and strains are known to a high degree of accuracy / Low weight is important (aircraft, aerospace, ...)
2 – 3	For less tried materials or Brittle materials under average conditions of environment, load and stress
3 – 4	For untried materials under average conditions of environment, load and stress or Better known materials under uncertain environment or uncertain stresses
8 – 10	When failure could cause serious injury or death (a steam boiler, pressure vessel, roller coaster, ...)
10 – 14	Unknown stresses in the manufacturing process (e.g. casting)

EXAMPLE 1.14



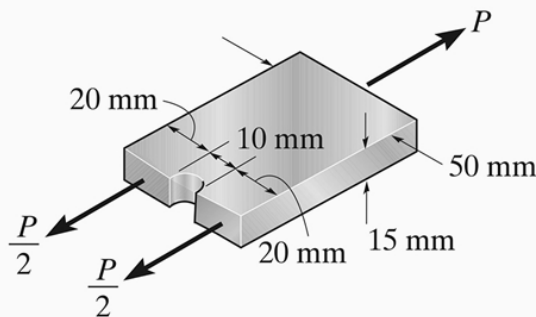
(a)

Determine the largest load P that can be applied to the bars of the lap joint shown in Fig. 1–27a. The bolt has a diameter of 10 mm and an allowable shear stress of 80 MPa. Each plate has an allowable tensile stress of 50 MPa, an allowable bearing stress of 80 MPa, and an allowable shear stress of 30 MPa.

SOLUTION

To solve the problem we will determine P for each possible failure condition; then we will choose the smallest value of P . Why?

Failure of Plate in Tension. If the plate fails in tension, it will do so at its smallest cross section, Fig. 1–27b.



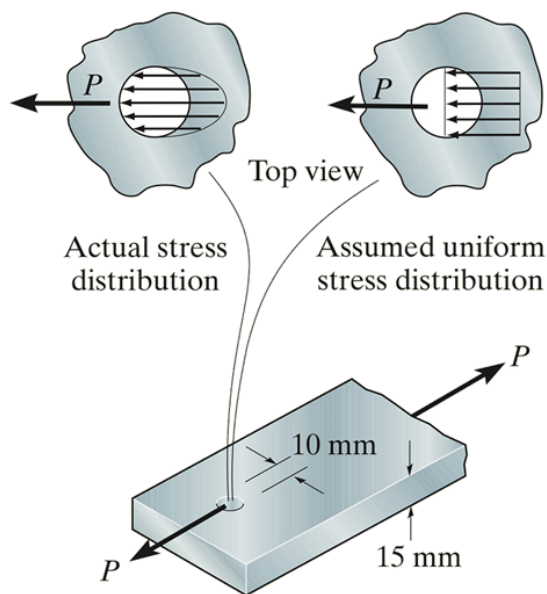
Failure of plate in tension

(b)

$$(\sigma_{\text{allow}})_t = \frac{N}{A}; \quad 50(10^6) \text{ N/m}^2 = \frac{P}{2(0.02 \text{ m})(0.015 \text{ m})}$$

$$P = 30 \text{ kN}$$

EXAMPLE 1.14 CONTINUED



Failure of plate in bearing caused by bolt
(c)

Fig. 1-27

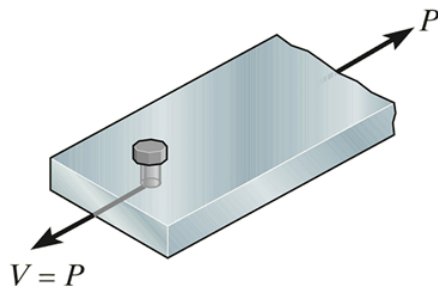
Failure of Plate by Bearing. A free-body diagram of the top plate, Fig. 1-27c, shows that the bolt will exert a complicated distribution of stress on the plate along the curved central area of contact with the bolt.* To simplify the analysis for small connections having pins or bolts such as this, design codes allow the *projected area* of the bolt to be used when calculating the bearing stress. Therefore,

$$(\sigma_{\text{allow}})_b = \frac{N}{A}; \quad 80(10^6)\text{N/m}^2 = \frac{P}{(0.01\text{ m})(0.015\text{ m})}$$

$$P = 12\text{ kN}$$

*The material strength of a bolt or pin is generally greater than that of the plate material, so bearing failure of the member is of greater concern.

EXAMPLE 1.14 CONTINUED



Failure of bolt by shear

(e)

Failure of Plate by Shear. There is the possibility for the bolt to tear through the plate along the section shown on the free-body diagram in Fig. 1-27*d*. Here the shear is $V = P/2$, and so

$$(\tau_{\text{allow}})_p = \frac{V}{A}; \quad 30(10^6) \text{ N/m}^2 = \frac{P/2}{(0.02 \text{ m})(0.015 \text{ m})}$$

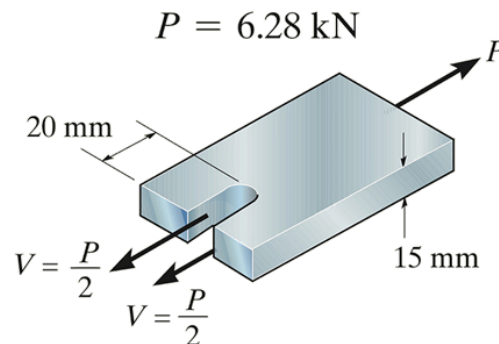
$$P = 18 \text{ kN}$$

EXAMPLE 1.14 CONTINUED

Failure of Bolt by Shear. The bolt can fail in shear along the plane between the plates. The free-body diagram in Fig. 1–27*e* indicates that $V = P$, so that

$$(\tau_{\text{allow}})_b = \frac{V}{A}; \quad 80(10^6) \text{ N/m}^2 = \frac{P}{\pi(0.005 \text{ m})^2}$$
$$P = 6.28 \text{ kN}$$

Comparing the above results, the largest allowable load for the connections depends upon the bolt shear. Therefore,



Ans.

Failure of plate by shear
(d)

Fig. 1–27 (cont.)

OUT-OF-CLASS READING

- **Textbook, pp. 46 – 54**